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This Is 25x More Effective Than Antidepressants —And It's Not a Drug

Installment nine in our series on understanding the truth about SSRIs.

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In this article we will examine a potential treatment for depression that so far looks to be almost 25 times more effective than antidepressants and four times more effective than ketamine at creating long-term psychological resilience.

Moreover, we will see that the impact of all three can be experimentally blocked in animals by blocking their impact on energy metabolism, which shows that the "psychiatric" model of how these drugs work to alleviate depression is wrong, and the "mitochondrial" or "energetic" model is correct.

This shows us that even when Pharma succeeds, it is flying blind and getting lucky.

The most important point, however, is that this nearly 25-fold (24-fold to be precise) better method is not a drug at all. It operates entirely outside the pharmaceutical model and is instead somewhat more akin to exercise, though the only fully natural way that our great-great-grandparents could have achieved it would have been to periodically travel into the mountains.

The core comparative study we will look at was done in mice, but we will put it in the context of the human research.

While randomized controlled trials are the best way to determine whether an intervention causes a benefit in humans, animal studies are extremely valuable in their greater ability to infer mechanisms. "Mechanism" is a fancy way to say "why and how something works." When you understand why and how something works, you can make much better decisions about how and when to use it as a tool. RCTs should be used to assemble what we know can work into our broader toolkit, while understanding how and why these tools work should be used to make wise decisions about what tool to use for the job.

The intervention we will look at is called **hypobaric hypoxia training.**

This is the opposite of hyperbaric oxygen therapy.

It is used to induce tolerance to low blood oxygen levels caused by low atmospheric pressure in, for example, military pilots.

We saw earlier in this series that serotonin's fundamental primary action is to act throughout the entire body to facilitate hypoxia tolerance, so it should not be surprising if we find that hypoxia is fundamental to depression.

Hypobaric hypoxia training can have adverse effects, including drowsiness that can onset hours after the training and could interfere with driving or operating machinery. If done without proper safety precautions, it carries all the same risks as altitude sickness. This often include transient headache, nausea, shortness of breath, an elevated stress response, getting sick, decreased cognitive performance, and lethargy while they more rarely can include delirium or even psychosis. Generally these respond to restoration of the oxygen level, so it would take some catastrophic misuse to generate the worst of these problems using a simulated altitude machine, whereas getting stuck in the mountains at a level of altitude you cannot tolerate can be harder to remedy quickly. Used properly, properly dosed hypoxia/altitude stress may be a safe and very powerful weapon to combat depression.

This article is not meant to suggest this as a first resort for depression — the basics of nutrient adequacy, a healthy lifestyle, and healthy mental practices should be the first resort — but rather to examine the fascinating prospects for this therapy and use it to properly understand the impact of SSRIs.

Nevertheless, I do believe that optimal health can best be produced with defined doses of hypoxic stress from periodic excursions to mountains or use of simulated altitude at sea level.

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Altitude, Hypoxia, and Depression

Ira Katz of the Albert Einstein College of Medicine in the Bronx, NY proposed in 1982 that hypoxia could cause depression by interfering with neurotransmitter metabolism. His evidence was largely limited to experimental studies showing the dependence of neurotransmitter regulation on oxygen and indications that depression might result in some medical conditions that cause hypoxia.

In 2018, authors from the University of Utah and the Salt Lake City Veterans Affairs Medical Center collected the evidence tying altitude to depression in the Harvard Review of Psychiatry. In the United States, rates of depression and suicidality are highest in the mountain states, and correlations adjusted for age, race, and sex suggest that a state's peak altitude explains 38% of its suicide rate while the altitude of its capital city explains 55% of its suicide rate. On a county basis, further controlling for income and population density, altitude explains 25% of the suicide rate. The effect size is large: high-altitude areas have a 3-fold higher rate of suicide than low-altitude areas. Similar results have been published for Spain, Saudi Arabia, South Korea, and in Peruvian electrical workers, though two studies failed to find associations in Turkey and India.

Six cases of new-onset panic disorder have been documented in travelers to the mountains of Nepal, anxiety has been found increased in mountaineers doing 55 days of simulated altitude training at the equivalent of 9000 meters, and depression and anger increased in US Marines doing simulated altitude training for thirty days at the equivalent of 2053 and 3600 meters.

Finally, chronic illnesses that cause hypoxia such as asthma and chronic obstructive pulmonary disease (COPD) cause more depression and suicide risk than other chronic illnesses. Both asthma and cigarette smoking are specifically associated with depression and suicide when *current*. Past asthma has no such association, and the association with cigarette smoking wanes during smoking cessation and returns if smoking is resumed.

With the benefit of 36 more years of research, these authors were able to tie hypoxia to brain energetics and relate it to the positive studies on creatine for depression, whereas Katz was operating in the emerging "neurotransmitter" model of the brain, which had been taking the psychiatry world by storm in the wake of the popular obsession with psychedelics in the 1960s and 70s.

Studies conflict about whether airline pilots have an increased or decreased risk of depression. Self-reported depression is higher among pilots, whereas official databases say they have less depression. My suspicion is that the databases are less accurate because pilots don't want to get taken off the job by seeking a diagnosis. Suicidal airline pilots have killed 562 people in crashes since 1972, so pilot depression is considered a security risk and the incentive is to hide it.

The blood cells of airline pilots have massively increased expression of the serotonin receptor. Stuck in the "psychiatric" model of serotonin as expected, these authors commented that "a dysfunctional serotonergic system leads to abnormal neurotransmission and the development of psychiatric disorders. Although our study was performed on peripheral blood cells, some authors have proposed that alterations in PBMC could be a reflection of alterations in the central nervous system (CNS)."

Thus they failed to appreciate that the massive increase in the expression of the serotonin transporter of **non-neuronal cells outside the brain** in pilots is showing the extreme significance of serotonin to the whole-body hypoxia response, not its role in mediating "positive vibes" within the brain.

Military pilots are often required to do hypobaric hypoxia training every few years, but its impact on depression is unstudied apart from the study of US Marines mentioned above. There are 25 papers that discuss the use of hypobaric hypoxia in airline pilots but they do not cover depression.

24 Times More Powerful Than Antidepressants

The central mouse study we will consider compared the antidepressant effect of hypobaric hypoxia training to the tricyclic antidepressant imipramine, with some side experiments that looked at ketamine.

The mice were subject to 70 days of chronic mild stress, including restraint, forced cold plunging, deprivation of food and water, tricking them with an empty water bottle, tilting their cage at a 45 degree angle, hours of strobe light exposure, and keeping them in a dirty, feces-full cage.

Multiple tests were used for depression and anxiety, including their preference for sugar, the degree to which they groom their fur, their ability to stay swimming when forced into a bucket of water instead of giving up, their willingness to explore an open area, and the ease with which they became afraid of loud noises.

They were split into four groups with treatments thrown into the middle of the chronic stress period: injection with imipramine or a saline control for four weeks, or exposure to four hours a day of hypobaric hypoxia equivalent to 5000 meters (16,400 feet) or a normoxic control for two weeks.

The difference in how long the treatments lasted is stunning:

- The antidepressant effect of imipramine faded away **one week** after treatment.
- The antidepressant effect of hypobaric hypoxia **lasted at least 12 weeks** after treatment.

The actual treatment shows an at least 12-fold longer-lasting effect of hypobaric hypoxia.

The imipramine was used for twice as long, so we could say from one perspective that it was 24-fold more powerful. However, the experiment was stopped there, so it lasts at least this long, and perhaps much longer.

They performed a number of more limited side experiments that showed the following:

- Blocking the hypoxia response at a molecular level abolished the antidepressant effects of hypobaric hypoxia training, imipramine, and ketamine.
- Chemically activating the hypoxia response by injecting cobalt chloride into the brainstem produced an antidepressant effect that lasted at least seven days.

More Powerful Than Ketamine?

This study showed that activation of the hypoxia response mediates ketamine's effects, but did not compare the robustness of the response to ketamine over time.

Nevertheless, there is a significant body of human literature on ketamine that we can compare this to.

Ketamine is a dissociative anesthetic. While it had been used for human surgery in the past it is now mostly used for veterinary surgery. It is used illegally and recreationally for its dissociative and hallucinogenic effects. Dissociation refers to detachment from emotional and physical reality.

A single infusion produces a powerful antidepressant effect, but its utility is limited by the fact that the doses required generally produce some degree of dissociation and often cause a rise in blood pressure. **The impact of a single infusion lasts for one to two weeks.**

Ketamine is primarily known as an NMDA receptor antagonist. Since schizophrenia involves NMDA receptor hypofunction, there is analogy between the ketamine-induced state and the schizophrenic state, which earns it a place in the category of "psychotomimetics."

Controversy exists over whether the dissociative effects mediate the antidepressant effects. If they do, this would fit into a "psychiatric" model of its impact rather than an "energetic" model because modulating neurotransmitter systems to achieve perceptual changes and psychological experiences would be the central mechanism of the drug.

On the one hand, most people experience some degree of mild dissociation during ketamine-based antidepressant treatment. On the other hand, while this does predict the antidepressant response, correlation coefficients indicate that it only predicts 4-17% of the response. This correlation is hardly evidence of causation. The dissociation could reflect a combination of the dose and distribution of the drug as well as the the sensitivity of the individual's receptors to it. That is, it could essentially be an index of the "individually-adjusted dose." The dissociative effect also represents a powerful capacity for a placebo effect. Regardless, 4-17% of the explanation of the variation is simply a weak proportion of the antidepressant effect. The perceptual changes are either entirely incidental to the antidepressant effect or play a minor role in it.

Given that its impact is blocked by molecularly blocking the hypoxia response in mice, the "mitochondrial" or "energetic" model is a much better explanation of the ketamine effect, just as it is for SSRIs and tricyclic antidepressants.

That the antidepressant effect of ketamine tends to last **one to two weeks** suggests **hypobaric hypoxia training could be 6-12 times more effective than it**, using the mouse study.

But this is based on single infusions.

There are isolated case reports of multiple ketamine infusions leading to remaining depression-free for six months to a year. However, larger studies show that this is not the normal result:

- In a study of 10 people receiving six infusions over two weeks, the antidepressant effect lasted **three weeks** after the treatment was stopped.
- In 11 people receiving 10-51 infusions over six to 49 weeks, four people continued on the maintenance regimen, four stopped because it stopped working, two stopped for unknown reasons, and one stopped because their 51st infusion caused a level of dissociation and restlessness that made them not want to use it again. This indicates that **you cannot maintain the effect perpetually because it will stop working *before* you stop using it in many cases, or you may run into a wall of adverse psychological effects.**
- In 15 people receiving six infusions over 12 days, one dropped out after the first infusion because it caused fatigue an irritability, one dropped out after the second because it didn't work, and one received no benefit even after the sixth infusion. Of the remaining twelve, nine achieved remission and three partially improved. In half of the twelve who got some benefit, **half were still benefitted four weeks out while the other half relapsed in an average of 16 days**. Nevertheless, many of them had declining neurocognitive and mental health scores between one and two weeks after treatment ended, and **everyone was in decline after the 2-week mark.**
- In 12 people receiving six infusions over two weeks and then four weekly followup infusions, five people dropped out before the initial six infusions were over. Two

withdrew because of adverse effects, one stopped because it wasn't working, and two withdrew because the ketamine was making their depression get worse. Two subjects partially improved and five achieved remission. In the four weeks that followed ketamine treatment, only one remained in remission. **The remitters all remained better four weeks out than they had been pre-ketamine, but they were also losing their peak benefit within one week after the last ketamine treatment.**

- In 100 people receiving six infusions over 12 days, 13 people dropped out during the infusion period and 10 more dropped out during the two-week followup. One dropout reported becoming suicidal and another dropout reported becoming manic. 26 achieved remission and 20 more had a positive response. **The antidepressant effect lasted robustly through the two weeks of followup.**

The most robust finding is that a ketamine infusion produces an antidepressant effect lasting one to two weeks. There is some indication that multiple infusions produce a longer-lasting effect that appears to max out in three to four weeks but be quite variable between individuals.

So **we can be generous to ketamine** and say that **hypobaric hypoxia training lasts three to four times as long** as a round of multiple ketamine infusions.

Many people are poor candidates for ketamine from the getgo. The ability to indefinitely sustain infusions to avoid this wearing off seems doubtful, and dragging them out over a year seems to often cause the ketamine to lose efficacy and sometimes become harmful.

Other Hypoxia Studies in Rats

The authors of the mouse study discussed above had conducted another earlier study in rats showing that intermittent hypoxia is similar in the magnitude of its antidepressant effect to Prozac and imipramine and that it lasts at least seven days after stopping the treatment, but did not compare its duration to that of the drugs.

A different group showed that in female rats but not male rats, altitude at 4500 feet or simulated altitude at 10,000 feet and 20,000 feet for one week caused a dose-dependent worsening on tests of depression and anxiety that were conducted at 4500 feet where the lab was situated. In another study this same group showed that female rats become anxious and anhedonic after altitude exposure.

The difference between these models is that the group showing benefits ran the tests a week *after* the hypoxia, whereas the group showing negative results seems to have run the tests immediately after removing the rats from their chambers.

If you view hypoxia stress similar to exercise, these findings are easy to reconcile. Right after exercising, your ability to complete the same task has gone down. Once you recover, your ability has gone up.

Intermittent Hypoxia Is an Essential Stimulus to Produce Positive Adaptations

The best way to synthesize all of this is that impaired oxidative energy production in the brain drives depression and anxiety, while SSRIs, tricyclic antidepressants, and ketamine are in variable ways helping to modify the response to this problem.

While the literature even on the antidepressant effect of ketamine is far more robust than the literature on the same effect of hypobaric hypoxia (where we only have rodent studies), and while the literature on SSRIs is yet more vast, the rodent studies on hypobaric hypoxia indicate it could be a vastly superior way to produce a long-lasting antidepressant effect.

There are hundreds of reasons that anyone at sea level could have impaired brain energy metabolism, but in people who live at altitude, chronic hypoxia is excessive and plays a major role in depression and suicide.

The increased risk of depression and suicide at moderate altitude coexists with a lower all-cause mortality risk, which shows that this degree of altitude stress is causing a tradeoff, showing the dose has some benefits but is not optimal.

While *chronic* altitude stress raises the risk of depression and suicide, *small doses* of altitude stress can improve the brain's ability to thrive under such stress, and thereby support psychological resilience.

In my review of the literature on simulated or real altitude raising testosterone in men, I concluded that 10-12 hours of 6500 feet using a simulated altitude tent three times a week is the best-performing protocol used in humans, which makes me hope that this can be consolidated to 5-6 days per month spent hiking, skiing, snowboarding, or doing whatever you want at the top of a mountain about 6500 feet above sea level.

In male boxers and cyclists, stacking the hypoxia stress for five or six days starts to cause the acute loss of the initial testosterone benefit, while spacing the protocol out three times a week allows the benefits to be maintained indefinitely, at least as far as has been studied. However, I believe that 5-6 days in the mountains would slightly push you into "overreach" from which you would bounce back more hypoxia-fit, testosterone-loaded, and psychologically resilient, while the rest of the month at sea level would allow you to reset your capacity to handle another round.

In non-athletes and recovering drug addicts, testosterone increases linearly during a month spent hiking the mountains of Tibet. While you could propose that natural mountain exposure is healthier than simulated altitude exposure, this cannot explain why depression and suicide is higher in people naturally living at altitude. Therefore, I believe the answer is that high-level athletic activity is already adding enough into the physical stress bucket that they become very sensitive to exceeding the optimal dose, whereas people who are sedentary are chronically physically understressed and can take longer to accrue the initial benefits.

Nevertheless, athletic activity is the vocation of everyone who has a body, as Nike cofounder Bill Bowerman once pointed out, so I think we should take the studies in boxers and cyclists as a provisional target for the optimal dose of hypoxia stress. This indicates that you want the equivalent of 15-20% of your time being spent at 6500 feet. We need more research to understand how changing the exact altitude changes the proportion of time that should be spent in it. For example, maybe we could spend two or three days per month at 12,000 feet.

To replicate the protocol of the human studies, you can use an altitude tent during sleep and part of the evening for 10-12 hours three times a week and can probably substitute a simulated altitude mask.

If you live at altitude, you may already have an excessive hypoxia response. If you cannot change where you live, you can take the opposite approach. Start using an oxygen machine at a low, practical dose such as an hour a day, and titrate it up to where you start seeing improved mood and psychological resilience. Then, continue titrating it slowly by increasing the dose a little each week. When you reach the point of diminishing returns, stay there or pull back a little. Hypothetically you would want to achieve something equivalent to sea level for 80-85% of your total hours, but you may find that the minimal dose producing the maximal effect you are seeking is lower than that.

In the first article in this series, we saw that exercise is a more effective antidepressant than Prozac in mice, while Prozac is an exercise-enhancing drug that increases muscle mass and grip strength.

We then saw that a central role of serotonin is to mediate the hypoxia response.

We built on this to see that SSRIs modulate this role of serotonin in promoting mitochondrial tolerance of oxygen deprivation and even enter the cell to have their own impacts on mitochondrial function.

We now see that intermittent hypoxia has a much longer-lasting antidepressant impact than any pharmaceutical drugs.

All of this converges on seeing depression as fundamentally about mitochondrial energy production and the impact of these drugs being far more about intersecting with mitochondrial energy production than modulating psychological effects of specific neurotransmitters.

In the remainder of this series we will look at how to approach depression without pharmaceuticals, how to deal with SSRI-induced mitochondrial dysfunction, and how to deal with mitochondrial dysfunction induced by SSRI withdrawal.

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Petrichor 1d

Liked by Chris Masterjohn, PhD

Walking long distance daily alleviated my depression and improved my overall health and sense of well-being. I really think walking is so underrated

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Ellymay 1d

Interesting to speculate on the effects of b12/iron deficiencies on depression via hypoxia. I had b12 deficiency many years with breathlessness and was also depressed. Now I have fixed my b12 deficiency I am no longer depressed.

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